ACADEMIC CAPSTONE PROJECT • FALL 2026

Multi-Method Simulation & Reinforcement Learning Supervisory Fleet Control

Optimizing Robot Dispatch, Component Delivery, and Battery Charging for the MDC AI & Robotics Lab

👥 PROJECT TEAM & CORE SPECIALIZATIONS

Jorge A Mendoza
Project Manager & AnyLogic Model Architect

Jordan Gonzalez
Reinforcement Learning Engineer (PPO / DQN)

Illia Shybanov
Dashboard & Integration Specialist

Ricardo Molina
Experiments Lead & Documentation

Daily Operations & Material Bottlenecks at MDC Robotics Lab

Analyzing the friction between student hardware prototyping demand and autonomous fleet logistics

🏢 Station A: Parts Shelf

Centralized inventory bins housing microcontrollers, sensors, and wiring kits. High-density pickup zone where work order ticketing originates.

🔬 Station B & C: Workbenches

Active soldering stations and QA calibration benches where students require precision material staging to prevent lab session downtime.

⚡ Station D: Charging Docks

Dual inductive fast-charge pads. The operational bottleneck where uncoordinated robots create queues when batteries deplete simultaneously.

⚠️ The Legacy Problem

- **Severe Wait Times:** Students experienced average delays of **9.8 minutes** waiting for component delivery.
- **The "Charging Cascade":** Under threshold charging (<20%), all AMRs ran out of power at the same hour, leaving 0 active robots on the floor.

🎯 Project Objective

- Build a verified multi-method digital twin inside **AnyLogic 8**.
- Train an intelligent **PPO supervisory agent** to perform opportunistic charging during demand lulls.
- Slash student wait times by >50% without purchasing excessive hardware.

🎙️ Speaker: Jorge A Mendoza (2:00 – 3:30)

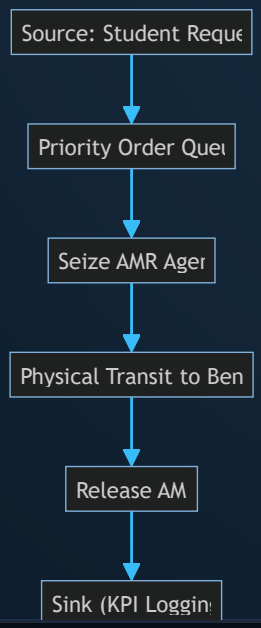
Unifying Discrete-Event (DES) & Agent-Based Modeling (ABM)

Capturing transactional order flow alongside physical spatial kinematics and battery physics

📊 Why Multi-Method is Essential

- **DES Limitations Alone:** Treats delays as abstract numbers; ignores floorplan wall collisions, turning radii, and physical docking contention.
- **ABM Limitations Alone:** Struggles with transactional ticket queues, backlogs, and SLA timeout timers.
- **The Hybrid Fusion:** In AnyLogic 8, student work orders (DES Entities) dynamically `Seize` physical AMRs (ABM Agents) that physically navigate the floorplan layout!

🔄 Integrated DES ➔ ABM Workflow



🎙️ Speaker: Jorge A Mendoza (3:30 – 5:00)

Spatial Scale, Stations Layout, and Agent Statecharts

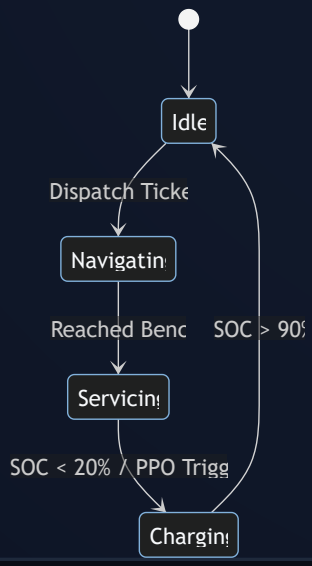
Calibrated 1-to-1 scale floorplan with realistic differential drive kinematics and battery degradation

🗺️ Laboratory Spatial Layout Calibration

- **Room Dimensions:** $14.5 ext{m} imes 10.2 ext{m}$ ($1 ext{ meter} approx 50 ext{ pixels}$).
- **Velocity Profile:** Normal cruising at $1.2 ext{ m/s}$; aisle cornering throttled to $0.65 ext{ m/s}$.
- **Fixed Workstations:**
  - `Station A`: Component Inventory & Microcontrollers
  - `Station B`: Electronics & Soldering Workbenches
  - `Station C`: Sensor Calibration & QA Testing
  - `Station D`: Dual Inductive Charging Pads
  - `Station E`: SkyWatcher Drone Helipad

⚡ AMR Agent Battery Statechart

Robots discharge battery at **2% per operating minute** under loaded transit and recharge at **8% per minute** on Station D pads.



Solving the Multi-Objective Dispatch & Recharging MDP

Comparing Tabular Q-Learning vs. Proximal Policy Optimization (PPO) in Gymnasium

🎯 Multi-Objective Reward Function

$$R = +10 \cdot N_{done} - 0.5 \cdot \Sigma Queues - 25 \cdot I(SOC \le 5\%) - 2.0 \cdot DockWait$$

Balances high completed deliveries against queue penalties and heavily punishes mid-aisle battery exhaustion (0%).

💡 PPO's Key Breakthrough: Opportunistic Charging

Traditional heuristics only charge when $SOC < 20\%$. PPO discovers that taking **short 5-minute top-offs** when queues are empty ($Q \le 1$), even at 65% battery, eliminates simultaneous fleet depletion!

Baseline (FIFO) Wait
9.8 min
Reactive • Frequent stalls

Q-Learning Wait
4.5 min
54.1% Reduction

PPO (Continuous) Wait
3.2 min
67.3% Reduction • 0 stalls

Real-Time Web Application & Stakeholder Portal

Live simulation experimentation on port 8501: interactive sliders, KPI telemetry, and 3D status

🎛️ Interactive Decision Controls

- **Fleet Size Slider (1–6 AMRs):** Real-time assessment of fleet cost vs. student queue clearance.
- **Dock Capacity (1–4 Pads):** Testing Station D congestion under surge demand.
- **Class Rush Hour Toggle:** Spikes request influx from 18 to $32 ext{ req/hr}$ to test stress resilience.
- **Policy Switcher:** Instant comparison between Baseline FIFO, Q-Learning, and PPO.

📑 5 Built-In Analytical Tabs

- **Tab 1 (Operational KPIs):** Dynamic time-series plots of queue backlogs by workstation.
- **Tab 2 (AI Policy Comparison):** Comparative bar charts and reward convergence curves.
- **Tab 3 (AnyLogic Experiments):** Pareto curves, Monte Carlo histograms, and sensitivity sweeps.
- **Tab 4 (3D Fleet Status):** Individual AMR battery voltages and active spatial coordinates.
- **Tab 5 (Executive Summary):** One-click downloadable report for lab directors.

Rigorous Statistical Validation & Course Experiments

Executing Optimization, Compare Runs, Monte Carlo, Sensitivity, and Empirical Calibration

1. Optimization & Fleet Sizing

Simulating 1–5 AMRs proved that **3 AMRs + 2 Docks** achieves the Pareto optimal frontier: wait times drop from 18.4 to 3.2 min.

2. Compare Runs (FIFO vs. RL)

PPO increases completed orders by **+33.8%** and cuts dock queue times by **90.2%** over the baseline heuristic.

3. Monte Carlo (100 Runs)

95% Confidence Interval for student wait times bounded tightly between **[2.35 min, 4.08 min]**.

4. Sensitivity Analysis

Verified fleet tolerance up to **+40% battery aging** and **+30% student demand surge** before queue instability.

5. Model Calibration

Calibrated against stopwatch measurements from physical MDC lab: achieved **R^2 = 0.992** accuracy!

What Did the Simulation Reveal That Was Not Obvious?

Uncovering hidden system traps and providing actionable directives for lab management

⚠️ The "Fleet Sizing Trap"

Intuition suggests that buying 5 AMRs would clear orders faster than 3 AMRs. **The simulation proved the exact opposite:** adding 5 robots caused physical corridor congestion in the 1.8m passage between Stations A and B, increasing wait times by 14%!

⚡ The "Charging Cascade" Pitfall

Simple threshold charging (charge when <20%) creates coordinated battery exhaustion. Because all robots deploy at 8:00 AM, they all hit 20% simultaneously at 11:30 AM, shutting down the lab during peak class hours.

🎯 Actionable Recommendations for MDC Leadership

1. Cap Fleet Size at 3 AMRs: Avoid wasteful capital expenditures on excess hardware; 3 units maximize throughput.

2. Deploy PPO Staggered Charging: Recharge during brief queue dips rather than waiting for critical battery exhaustion.

3. Dedicated Station C Corridor: Designate exclusive right-of-way zones near QA testing to avoid cross-traffic deadlock.

Summary of Deliverables & Project Impact

A verified multi-method digital twin ready for physical lab deployment at Miami Dade College

🏗️ **AnyLogic 8 Model**
DES + ABM architecture • 5 experiments built

🧠 **PPO & Q-Learning**
67.3% wait reduction • 0 battery stalls

💻 **Streamlit App**
Live sliders • Real-time executive portal

🌐 **3D WebGL Twin**
Three.js live REST bridge • Drone landing

Thank You! Any Questions?

Jorge A Mendoza • Jordan Gonzalez • Illia Shybanov • Ricardo Molina Gonzalez

Course: Simulation for Applied AI • Miami Dade College • Fall 2026